NACAD: A Noise-Adaptive Context-Aware Detector for Remote Sensing Small Objects

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***Abstract*— Small object detection in remote sensing faces significant challenges, such as their offset sensitivity caused by the small area coverage, the dim targets in images, and their vulnerability to complex backgrounds, which often result in missed detections and false alarms. In this work, we propose a noise-adaptive context-aware detector (NACAD) to alleviate the above problems, which mainly consists of a region proposal network (RPN) with noise-adaptive module (NAM), a context-aware module (CAM), and a position-refined module (PRM). The main contributions are threefold. First, we leverage the information around small objects as positive-incentive noise (also known as π-noise); through enlarging the range of small objects by NAM, more anchors of them are preserved as positive samples, thus stimulating the model to detect small objects. Second, the CAM is designed to provide multiple observation perspectives and abundant contextual representations for the enhancement of object features. Third, to reduce the interference of pure noise in the complicated backgrounds around small objects, the spatial calibration along two coordinate axes is devised by PRM to optimally use information beyond object regions. The effectiveness of our proposed detector, particularly on small objects, has been validated by the experiments on two public datasets, ITCVD and HRRSD. In particular, the NAM improves the recall of small objects, CAM enhances small object features, and PRM helps address the pure noise in complicated backgrounds around small objects.**

***Index Terms*— Context aware, noise adaptive, offset sensitive, remote sensing, small object detection.**

I. Introduction

RECENTLY, with the developments of deep learning techniques, the object detection approaches for high-spatial-resolution remote sensing (HRS) images have met a great progress [1], [2]. As one of the most hot-topic problems of HRS image processing, HRS object detection is a crucial research field and has been applied in various fields, such as agriculture, transportation, and security.

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For instance, in the road congestion analysis task, vehicles are detected and counted for transportation planning or intersection traffic control [3]. However, HRS object detection usually focuses on city-scale information extraction, which requires HRS images of a very large range, such as a whole urban area. As shown in Fig. 1, it can be tough to detect small objects, such as vehicles, from such large-range HRS images. The need for detecting small objects is also present in other HRS object detection applications, such as sea rescue [4], autonomous driving [5], and pedestrian detection [6]. The difficulties of small object detection will greatly limit applications of HRS object detection in reality. Therefore, the small object detection has emerged as a hot research topic and been full of challenges for remote sensing images.

A widely accepted definition of small objects is provided by the microsoft common objects in context (MS COCO) metrics, which define small objects as those that occupy an area of less than 32×32 pixels [7]. Significant studies have been conducted to detect small objects, such as ships, vehicles, airplanes, storage tanks, and traffic signs in HRS images. For instance, Zhang et al. [8] introduce an anchor-free ship detector named GRS-Det, which employs a rotation Gaussian mask to accurately represent ships and integrates a selective concatenation module to blend multilevel features. Li et al. [9] propose a rotation-insensitive and context-augmented detection framework simultaneously. Their framework incorporates a network for fusing local and contextual features, which enhances the small object detection. Besides, Lu et al. [10] develop a novel GACLNet that enhances localization performance and has been shown to be effective for detecting small objects.

However, the aforementioned approaches ignore the adverse effects caused by the characteristics of HRS small objects, which include the offset sensitivity triggered by their small area coverage, the dim targets in images, and the vulnerability to complex backgrounds. In dealing with these issues, a novel noise-adaptive context-aware detector (NACAD) is proposed for the small objects in HRS images, which is a typical application of π-noise [11].

First, a novel noise-adaptive module (NAM) is proposed to prevent the small object anchors from being lost in the region proposal stage. Specifically, through enlarging the range of small objects, more small objects containing helpful information are preserved as positive samples, which is beneficial to improve recall performance. Second, a context-aware module (CAM) is composed of four densely connected 3×3

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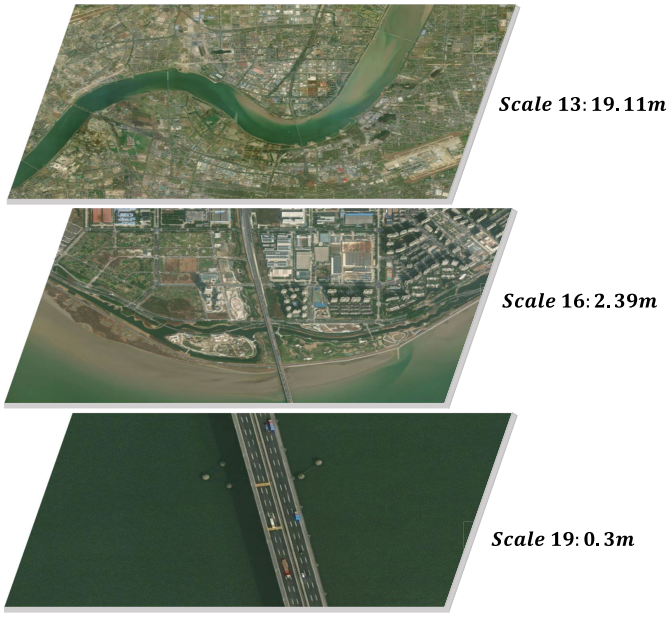


Fig. 1. Different scales of HRS images. Label # in "*Scale #*" refers to the image scale level. Spatial resolution of each scale level is behind "*Scale #*."

convolutions, each with a different dilation rate, and serves to amplify the signals of HRS small objects, thereby making their features more distinctive and easier to recognize. Third, a position-refined module (PRM) is designed to optimally leverage information beyond small object regions. Because of spatial calibration along the x-axis and the y-axis, it filters out the pure noise in the complex background around small objects, thereby mitigating interference with detection results.

In summary, we present a novel detector specifically designed to overcome the challenges associated with HRS small object detection, arising from their inherent characteristics. Experiments demonstrate that our method gains substantial improvements on the HRRSD dataset (10.7% $\boldsymbol{AP_S} \uparrow$ and 15.4% $\boldsymbol{AR_S} \uparrow$). The primary contributions of this article are as follows.

1) To prevent the small object anchors from being lost in the candidate region generation part, the background around small objects are introduced as π-noise by NAM, which is simple and effective in reducing the omission of small objects during detection.
2) To strengthen the signal of HRS small objects and make their features more discriminative, a compact and efficient CAM is introduced. This module adeptly leverages contextual features, thus significantly reducing the incidence of false alarms in the detection of small objects.
3) To address the issue of aliasing in background information, the PRM is presented to effectively filter out pure noise from small object backgrounds, eliminating its interference with the detection results, which helps improve the accuracy and reliability of detecting small objects in complex backgrounds.

The rest content is organized in the following manner. Section II reviews the related works of general object detection paradigms, small object detection methods, π-noise, and context information for small object detection. Section III provides a detailed introduction of the NACAD, while Section IV presents and analyzes the experimental findings. Finally, the conclusion is drawn in Section V.

II. Related Works

A. General Object Detection Paradigms

Benefiting from the feature extractors and discriminators that are more adaptive for different tasks and object categories, current object detectors are extensively based on deep CNNs. They can be classified into two types, namely, two-stage detectors and one-stage detectors, based on their prediction stages.

1) Two-Stage Object Detectors: These approaches typically involve predicting potential bounding boxes (BBoxes) for objects in the initial stage and refining them as well as predicting their categories in the second stage. R-CNN [12] is a groundbreaking object detector that introduced the use of CNN features for the first time. Soon, shared feature map and region of interest (RoI) pooling improve the training speed, resulting in Fast R-CNN [13]. Then, Faster R-CNN [14] becomes a widely used end-to-end framework for object detection, by incorporating the almost cost-free RPN, which generates region proposals effectively. Later, a position-sensitive RoI pooling layer was introduced by R-FCN [15] to balance between accurate localization and translation invariance in object detection. Also, Cascade R-CNN [16] refines object BBoxes gradually to suppress the close false positives, achieving significant performance improvements.

2) One-Stage Object Detectors: In contrast, for one-stage object detectors, both the object categories and their BBoxes are predicted directly, without the need for a separate proposal generation step. The OverFeat [17] slides the fully convolutional network (FCN) [18] over an input image as a feature extractor to detect objects from pixel to pixel. Then, YOLO [19], [20] takes as input the entire image, and the localization and classification results are directly predicted at the output layer. SSD [21] generates BBoxes by using a set of default boxes with varying aspect ratios and scales, ensuring diversity in the anchor boxes. Besides, multiscale feature maps are adopted in this network. Furthermore, RetinaNet [22] devises the focal loss, which enhances the standard cross-entropy criterion by assigning weights to the hard examples during training and suppressing the easy negatives. This method aims to address the issue of extreme foreground–background class imbalance. Recently, several anchor-free methods, such as CornerNet [23], CenterNet [24], and FCOS [25], have emerged, which replace the anchor mechanism with key points or central points.

B. Small Object Detection Methods

Current methods for detecting small objects mainly focus on improving detection by addressing the issue of insufficient information content. These methods usually introduce tailored components to general object detection paradigms, which are summarized comprehensively in the recently published review [26]. Kisantal et al. [27] focus on increasing the number of small objects by oversampling images so as to

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relieve the object-scale imbalance in datasets. Considering the insufficient appearance evidence, Liu et al. [28] leverage super-resolution methods, such as generative adversarial networks (GANs), to obtain better feature representations of small objects [29]. Besides, Liu et al. [21] detect multiscale objects in a divide-and-conquer fashion using corresponding-level representations, in order to avoid the disappearance of small object feature response. Furthermore, Lin et al. [30] propose a feature-fusion approach that integrates features from different layers [31] to obtain discriminative representations, which include both spatial and semantic information for small objects. Wang et al. [32] introduce a new evaluation metric named normalized wasserstein distance (NWD), which utilizes 2-D Gaussian distributions of BBox to overcome the sensitivity of intersection of union (IoU)-based metrics to small object location deviation and has demonstrated encouraging performance.

C. Positive-Incentive Noise

Positive-incentive noise is an inspiring concept first proposed by Li [11]. The author revisits the impact of noise and finds that noise does not always result in a negative impact. Stochastic resonance [33] is a common approach that utilizes random noise to improve the detection of weak signals. The concept of task entropy, which is the complexity of a task, can be used to classify traditionally defined noise into two categories: π-noise and pure noise. π-noise refers to random signals that simplify the target task. Extensive experimental evaluations on various fields demonstrate the effectiveness and universality of π-noise. So, in this article, we introduce the surrounding area of objects as π-noise to facilitate the selection of small object training samples.

D. Context Information for Small Object Detection

From an information theory viewpoint, incorporating various features can boost machine learning performance, as Zhang et al. [34] have highlighted. Different features offer complementary insights, enhancing data representation. Zhang et al. present MFR-CNN, a traffic object detection method merging multiscale features with global context. Context information includes prior knowledge, such as semantic and spatial relationships. Leveraging associations between objects and their surroundings, or between objects, enhances object recognition. Models integrating this context often outperform baselines, particularly for small objects confined to limited image regions.

Zhang et al. [35] introduce a strategy for effectively utilizing foreground information to tackle high background complexity in remote sensing images. Their method, ForDet, injects foreground-contextual representations and highlights the potential foreground anchors to boost the detection of foreground areas. This method has good performance in small objects, such as ships and cars. Chen et al. [36] incorporate the features of RoI and its surrounding context region, with both regions with a coinciding center for subsequent detection. Similarly, Gong et al. [37] also leverage the underlying context information surrounding the RoIs, which is obtained through a context-RoI mining layer. For each RoI, the corresponding context RoI is defined as the minimum BBox that encompasses both the RoI and several context proposals. Our approach differs from aforementioned methods in two ways: 1) we enlarge the area scopes of small objects to retain more small object anchors, while they leverage the contextual region to acquire more informative features and 2) we introduce the background information only around small objects (less than 32×32 pixels), while they extract surrounding features for all objects.

III. Methodology

Our NACAD builds upon the Faster R-CNN baseline, incorporating feature pyramid network (FPN) with CAM, RPN with NAM, and RoI head with PRM, as depicted in Fig. 2. In this section, we start with a brief review of the foundational Faster R-CNN model; then, we detail the three components, systematically addressing their respective problems, principles, and methods. Finally, we explain the overall loss function of this detector.

A. Revisiting Faster R-CNN

As one of the most cited works, Faster R-CNN [38] is taken as a classical object detection framework. Main parts of this framework are as follows: image feature generation, candidate region generation, RoI pooling, localization, and classification. The illustrations of different parts are as follows.

1) The image feature generation part takes CNNs to extract a feature map of the entire image, from which different object features are later sampled during processing.
2) Given the whole image feature map, the candidate region generation part takes several convolutional layers to predict candidate object regions, i.e., RoIs. First, at each pixel, there are predefined BBoxes with different areas and aspect ratios. Then, the convolutional layers are used to predict location regressions and classes. Finally, the predefined BBoxes are filtered by the predicted classes and fine-tuned with the predicted location refinements, to get the candidate object regions.
3) The RoI pooling part samples candidate object regions from the whole image feature map to get the candidate object region feature map. To be noticed, the sampling actually uses max or mean value to represent a region, and the candidate object region feature map is fixed on size.
4) The classification part takes several convolutional or fully connected layers to predict the class probability vector, which is then converted into a one-hot vector. The vector has a single element with a value of 1 and all others with a value of 0.
5) The localization part takes several convolutional or fully connected layers to predict a coordinate refinement vector with C times the length of a BBox vector of 4. The ratio C represents the number of object classes.

As the most classic two-stage network, Faster R-CNN maintains high detection accuracy, and many algorithms build upon it. However, for the detection of small objects in HRS images, the network still has some points to improve.

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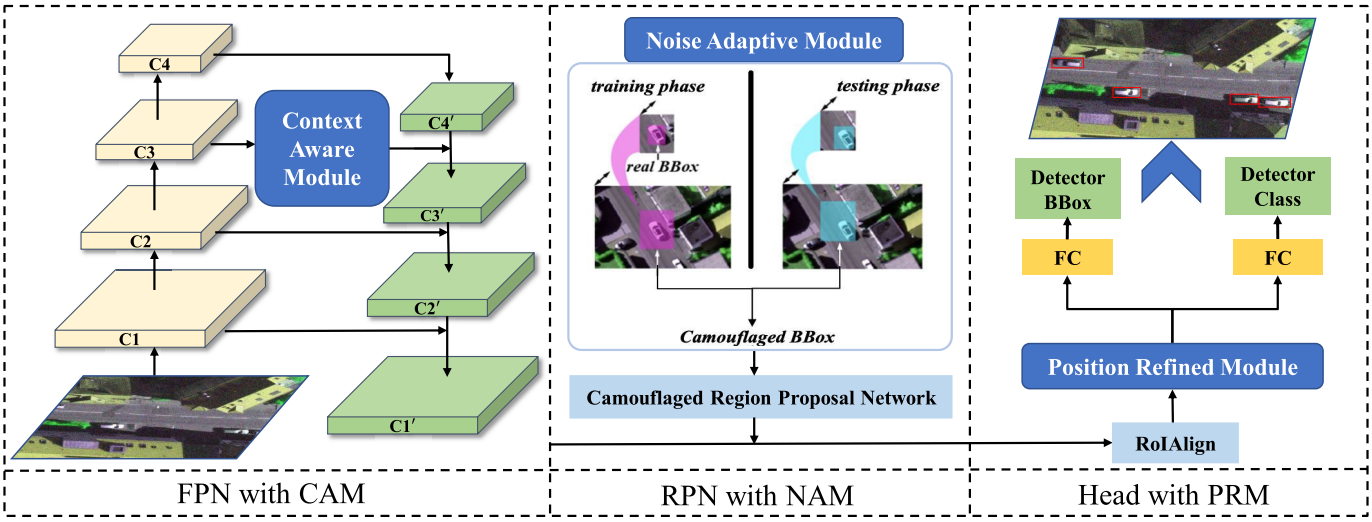


Fig. 2. Illustration of the proposed NACAD.

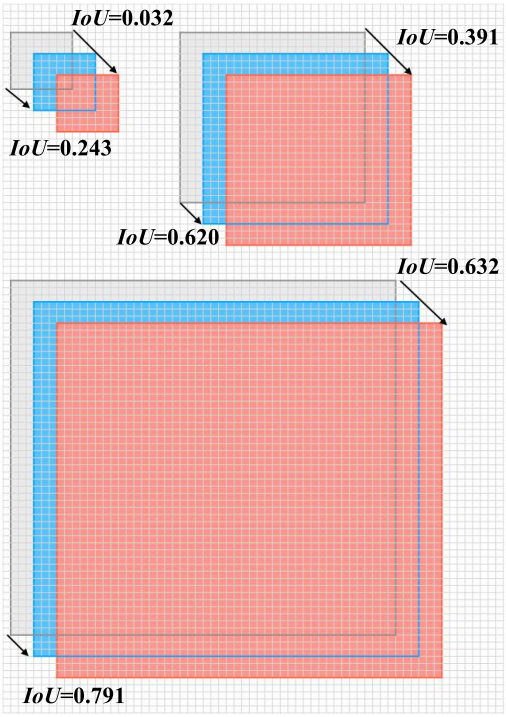


Fig. 3. Offset-sensitive problem of HRS small objects. The small, medium, and large objects (16 × 16, 48 × 48, and 100 × 100 pixels, respectively) are denoted by top-left, bottom-left, and right positions, with each grid representing two pixels. For boxes of three scales, gray represents ground truth of objects, and blue and pink represent predicted boxes that offset 6 and 12 pixels, respectively, along the diagonal direction.

B. Noise-Adaptive Module

1) Problem: Given the small coverage area of HRS small objects, even a minor positional deviations in anchors from ground truth can significantly lower IoU scores. Referring to Fig. 3, it can be observed that a small object experiences a substantial drop in IoU (from 100% to 24.3%) due to a slight displacement of the anchor box (six pixels along the diagonal direction), whereas the medium and large objects encounter comparatively smaller drops (62.0% and 79.1%, respectively). Meanwhile, a greater deviation (12 pixels) results in more serious IoU decline of small objects (from 100% to 3.2%).

In the candidate region generation part, the area overlap ratio between an anchor (A) and a real BBox (RB) is always used to select positive samples for training, which can be defined as follows:

$$\mathrm{IoU}_{\mathrm{candidate}} = \frac{A \cap \mathrm{RB}}{A \cup \mathrm{RB}}. \tag{1}$$

For all anchors, those whose $\mathrm{IoU}_{\mathrm{candidate}}$ is higher than 0.7 with respect to ground truth BBoxes are labeled as positive samples (i.e., containing objects of interest), while those with an IoU lower than 0.3 are regarded as negative samples (i.e., background regions).

Affected by the offset-sensitive attribute of small objects, small-scale anchors are more vulnerable to slight offset, resulting in a large drop of $\mathrm{IoU}_{\mathrm{condidate}}$, which means that they are more likely to be regarded as negative samples or eliminated. Less small-scale positive samples will reduce the role of small objects in model training, thus damaging the model's detection ability to small objects, resulting in low recall performance for these objects. One reason for ineffective detection by the model is the inadequate representation of small objects in the training data, which can impact the model's ability to accurately detect and classify such objects in real-world scenarios.

2) Principle: The working principle of the proposed NAM is analyzed from the perspective of locating and recognizing small objects. For the locating of small objects, the classical framework is likely to lose small-scale RBs. In a sharp contrast, we consider introducing background information around small objects as π-noise, which takes a larger scale BBox to camouflage the original small-scale BBox. Because of this operation, the proposed network has larger localization probability and smaller localization difficulty.

What about the recognition of small objects? For an input RoI containing a real object, there are three types of information: relevant objects information I_{ob}, relevant backgrounds information I_{bg}, and irrelevant backgrounds information I_{ir}.

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In general, objects information I_{ob} is relevant to the object category recognition, and backgrounds information is irrelevant to the object category recognition. Despite, CNNs can detect patterns even in background pixels within the BBox. In the network training phase, real object pixels and background pixels in the RoI BBox are used for learning a network model. While learning recognition with the background pixels inside the RoI BBox, the network model can find a latent pattern between recognition and the object context background. This latent pattern is named relevant background information I_{bg} in this article. Then, the rest backgrounds are irrelevant backgrounds information I_{ir}.

Then, the information of an RoI I_{RoI} can be formulated as follows:

$$I_{RoI} = S + N \tag{2}$$

where the signal S and the noise N are formulated as follows:

$$S = I_{ob} + I_{bg} \tag{3}$$

$$N = I_{ir}. \tag{4}$$

Finally, the signal-to-noise ratio (SNR) is

$$\text{SNR} = \frac{S}{N} = \frac{I_{ob} + I_{bg}}{I_{ir}}. \tag{5}$$

According to the property of the SNR, SNR is negatively relevant to the object recognition difficulty.

Considering the small object, the low detection accuracy comes from too much backgrounds information and too less object information and results in a large I_{ir} as well as a small I_{ob}. Thus, the SNR will be too small for effective recognition. In other words, the positive object recognition probability P_{rec} is positively correlated with the object size a

$$P_{rec} \propto a. \tag{6}$$

3) Method: To improve recall of small objects, the π-noise is introduced to alleviate the offset-sensitive problem. As shown in middle block of Fig. 2, compared with the classical framework aiming at predicting RBs by RPN, the proposed NAM aims at the prediction of camouflaged BBoxes. The camouflaged BBox is explained as follows.

In the training phase, given an RB (x_c, y_c, w, h), our proposed method expands the small object BBox labels by k times to get the camouflaged BBox (x_c, y_c, kw, kh), where (x_c, y_c) and (kw, kh), respectively, denote the center point and the BBox size. π-noise is contained in the extra information of camouflaged BBox. It is noteworthy that small objects are those with fewer than 32×32 pixels. While in the testing procedure, the proposals produced from RPN are also camouflaged BBoxes with lager regions.

C. Context-Aware Module

1) Problem: In HRS imagery, small objects often manifest with muted signal responses. The various geometries of these objects, combing with their slight color variances, lead to features that are not just less prominent but also present significant challenges in terms of differentiation.

2) Principle: Since signals from small objects are typically weak, incorporating context information as auxiliary information can significantly enhance their feature maps. Hence, a novel CAM is embedded in the FPN by combining dilated convolution groups and dense connections to provide multiple observation perspectives and increase the interactions among them.

We compute multiple dilated convolutions [39] groups on a single branch to obtain several larger receptive fields, which are interrelated with each other by dense connections. The receptive field of our CAM is 41, which is lager than the commonly used size 9. Larger receptive field has broader perspective, containing more spatial relationship information about objects. Connections from previous feature groups to the posterior layers can inherently facilitate the interaction between multiple contextual information, thus increasing the interaction between observation perspectives. For instance, the last group has various receptive fields, including 1, 5, 13, 25, and 41. Thus, global and local context information can be combined completely to supplement the features of HRS small objects.

Besides, the larger the receptive field, the more it can correspond to the features of larger scale objects. Due to our CAM possessing multiple receptive fields, it is capable of enhancing the features of objects across various scales.

3) Method: As shown in Fig. 4, the CAM consists of four sets of dilated convolutions in series, which greatly expand receptive field of original feature map. Each set consists of one 1×1 convolution, one 3×3 convolution with different dilation rates, and one 1×1 convolution, denoted by $\mathbf{D}_i$, where $i \in \{1\text{–}4\}$. The output of the $\mathbf{D}_i$ is defined as $\mathbf{F}_i \in \mathbb{R}^{C\times H\times W}$, and they have the same dimensions as the original input feature map $\mathbf{F}_0 \in \mathbb{R}^{C\times H\times W}$. d means the dilation rate of the middle 3×3 convolution, and the dilation rates of four 3×3 convolution are 2, 4, 6, and 8. As indicated at the bottom of Fig. 4, the input feature has a receptive field of $r_0 = 1$, and the receptive fields of four sets convolutions are $r_1 = 5$, $r_2 = 13$, $r_3 = 25$, and $r_4 = 41$, respectively. The two 1×1 convolutions are used inside the dilated set to increase feature interaction.

Besides, dense connections are adopted in the proposed CAM. First, output feature maps of $\mathbf{D}_{i-j}(\cdot)$ are added to the output of $\mathbf{D}_{i-1}(\cdot)$ by elementwise summation. Then, they are fed into $\mathbf{D}_i(\cdot)$ to mine more contextual information. To be more specific, the function $\mathbf{F}_i(\cdot)$ can be expressed in the following manner:

$$\mathbf{F}_j = \mathbf{D}_j\left(\sum_{k=0}^{j-1}\mathbf{F}_k\right), \quad j \in [1, 4] \tag{7}$$

where the sum method is elementwise summation.

Each set of dilated convolution feature offers a unique perspective due to its distinct receptive field. Thereby, the proposed CAM effectively leverages these contextual features to amplify the responses of HRS small targets while fostering greater interaction between these perspectives.

D. Position-Refined Module

1) Problem: First, as we can see in Fig. 1, in HRS images, the presence of complex and highly similar background

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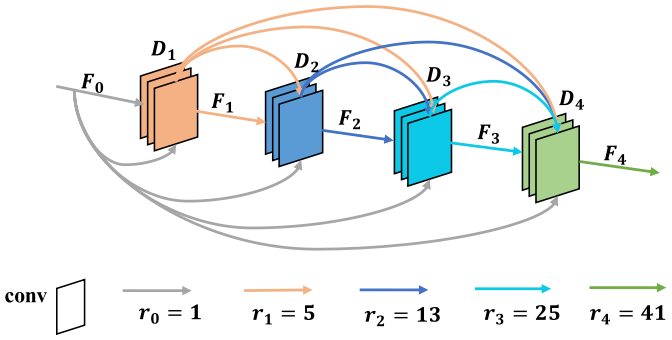


Fig. 4. CAM.

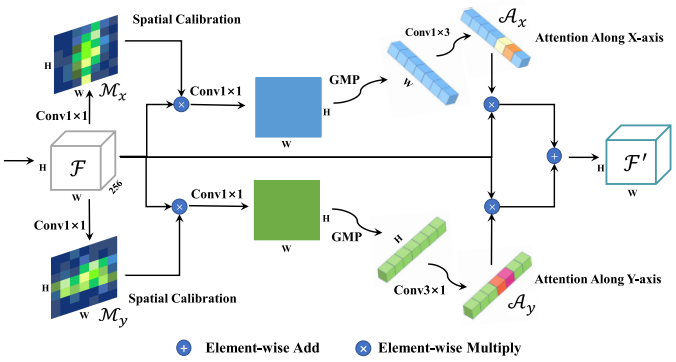


Fig. 5. PRM.

elements adds a layer of difficulty to small object detection. These overlapping and closely resembling elements in the backdrop can often create a visual noise, causing confusion in the differentiation process. In addition, benefiting from the π-noise, the proposed NAM can improve the recall of small objects through introducing informative background information. However, the background information includes not only π-noise but also pure noise. Especially, when the camouflaged BBox contains other RB, such as the camouflaged BBox of dense small objects, the extracted features are even less characteristic. So, for small objects in the RoI with surrounding background information, how to increase the network's ability to distinguish pure noise from π-noise?

2) Principle: Recently, there are several works devoted to incorporate spatial attention or channel attention to advance the performance of CNNs [40], [41], [42]. To refine the feature map and selectively focus on salient parts, we design a PRM inspired by Wang et al. [43]. Different from the common attention mechanisms, the proposed PRM extracts attention weights along the x-axis and y-axis directions, thus reducing the interference of pure noise in the complicated backgrounds around small objects.

3) Method: In order to achieve reasonable use of π-noise, the PRM is introduced in the detection head to eliminate pure noise mixed in the background. As shown in Fig. 5, it first performs spatial calibration utilizing the $k \times k$ RoI features $\mathcal{F}$ ($k = 7$), which are obtained by RoIAlign [15] and then determine the specific location of small objects from two directions, respectively.

Specifically, two 1×1 convolutions are first employed to produce attention maps $\mathcal{M}_{\mathbf{x}}$ and $\mathcal{M}_y$. Through the elementwise multiply between RoI feature and attention maps, we enhance the RoI feature from horizontal and vertical directions, respectively. After obtaining the attention maps $\mathcal{M}_x$ and $\mathcal{M}_y$, aggregation is performed on $\mathcal{F}$ to derive $\mathcal{A}_x$ and $\mathcal{A}_y$ using the following procedure:

$$\mathcal{A}_x = f_2(\text{GMP}(f_1(\mathcal{F} \otimes \mathcal{M}_x))), \quad \mathcal{A}_x \in \mathbb{R}^{1\times k} \tag{8}$$

$$\mathcal{A}_y = f_2\big(\text{GMP}\big(f_1\big(\mathcal{F} \otimes \mathcal{M}_y\big)\big)\big), \quad \mathcal{A}_y \in \mathbb{R}^{k\times 1} \tag{9}$$

where f_1 is the 1×1 convolution layer to reduce feature channels and f_2 is the 1×3 or 3×1 convolution layer to refine feature vector. $\otimes$ means elementwise multiply, and global max-pooling (GMP) can be expressed as follows:

$$X_{\max} = \text{GMP}(X) = \max \sum_{i=1}^{H} \sum_{j=1}^{W}, \quad X_{\max} \in \mathbb{R}^{1\times k} \tag{10}$$

where H and W are the height and width of the feature map, respectively. Then, the original feature $\mathcal{F}$ is multiplied by $\mathcal{A}_x$ and $\mathcal{A}_x$, respectively, to establish attentions along the x-axis and the y-axis. Finally, both attention features of horizontal and vertical directions are integrated together, which can be written as follows:

$$\mathcal{F}' = (\mathcal{F} \otimes \mathcal{A}_x) \oplus \big(\mathcal{F} \otimes \mathcal{A}_y\big) \tag{11}$$

where $\mathcal{F}'$ is the output feature of PRM and $\oplus$ means elementwise add.

Therefore, by refining the feature maps and selectively focusing on salient regions, PRM effectively mitigates interference from the HRS background pure noise. This module allows for the optimal utilization of information beyond small object regions.

E. Loss Functions

The network's global loss function is composed of individual losses for the RPN (with NAM), classifier, and regressor, aimed at optimizing the object detection performance, and is defined as follows:

$$L = L_{c-\text{RPN}} + L_{\text{CLS}} + L_{\text{REG}} \tag{12}$$

where $c-$RPN represents RPN with camouflaged BBoxes output. To be noticed, the BBox regressor in RPN is trained with camouflaged BBox labels $\widetilde{B}_{\text{gc}}(\widetilde{x_c}, \widetilde{y_c}, k\widetilde{w}, k\widetilde{h})$, which are k times the size of original BBox labels. Taking u as the real object category index, the RPN with camouflaged BBox loss term is formulated as follows:

$$L_{c-\text{RPN}} = L_{c-\text{rec}} + \lambda_c[u \geq 1]L_{c-\text{loc}} \tag{13}$$

$$L_{c-\text{rec}} = \text{SoftMax}\big(p_{\text{gc}}, u\big) \tag{14}$$

$$L_{c-\text{loc}} = \text{Smooth}L_1\big(B_{\text{gc}} - \tilde{B}_{\text{gc}}\big) \tag{15}$$

where $\lambda_c[u \geq 1]$ equals 1 if $[u \geq 1]$ and 0 if else. The functions of SoftMax and SmoothL_1 are formulated as

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follows:

$$\text{SoftMax}\left(p_{\text{gc}}, u\right) = -\log \frac{e^{p_u}}{\sum_c e^{p_c}} \tag{16}$$

$$\text{Smooth}L_1(x) = \begin{cases} 0.5x^2, & \text{if } |x| < 1 \\ |x| - 0.5, & \text{otherwise.} \end{cases} \tag{17}$$

The remainder of the loss function is identical to the definition provided in the Faster R-CNN framework.

IV. Experiments

This section presents a range of experiments to quantitatively and qualitatively evaluate the NACAD method, along with detailed descriptions of the datasets, evaluation metrics, and experimental settings.

A. Datasets

To demonstrate the efficacy of our small objects detector, experiments are performed on two widely used public datasets.

1) ITCVD Dataset [44]: It is a large-scale vehicle detection dataset that has 173 images with a size of 5616 × 3744, containing 29 088 vehicles. In this article, we cut one image into 16 images of 1404 × 936 for training purpose. Then, we redivide the training set, test set, and verification set for experiment, which account for 25%, 25%, and 50%, respectively. It is noteworthy that in the training set, the smallest object size is 18 × 15, while in the test set, the minimum target size is 19 × 10. Besides, 46.9% of the object annotations in the training set have a height or width less than 32, which are enlarged by NAM during training. Since nearly half of the training samples are small objects, ITCVD is sufficient to verify the effectiveness of our proposed NACAD on small objects.

2) HRRSD Dataset [45]: The HRRSD dataset contains 55 740 object instances belonging to 13 different categories, with approximately 4000 instances per category. The dataset comprises 21 761 HRS images with spatial resolutions ranging from 0.15 to 1.2 m, which are obtained from Google Earth and Baidu Map. The data are devided into three sets, with the training, validation, and test sets comprising 25%, 25%, and 50% of all images, respectively. There are abundant categories of objects in the HRRSD dataset, while including various small objects, such as storage tanks, ships, vehicles, tennis courts, and airplanes, all with a minimum geometric mean size less than 32. Notably, the storage tanks and ships possess minimum geometric mean sizes of 22 and 23, respectively. This rich multiclass dataset stands as an ideal benchmark for assessing the efficacy of our approach in scenarios that more closely reflect real-world conditions.

B. Evaluation Metrics

PASCAL visual object classes (VOC) metric [46] and MS COCO metric [7] are two kinds of evaluation metrics being used universally to measure the detection accuracy. In this article, the MS COCO evaluation criterion that employs more comprehensive evaluation metrics is leveraged to evaluate our proposed small object detector. Especially, we focus on $\boldsymbol{AP_S}$ and $\boldsymbol{AR_S}$ for small objects. Besides, to evaluate the model complexity and inference efficiency of various methods, we consider floating point operations per second (FLOPs), parameters (Params), and frames per second (FPS) as key metrics.

There are ten metrics used to compare the detection performance, which reflect the object detection performance at various IoU thresholds and for objects of different scales. Among them, the metrics calculated based on IoU threshold include $\text{AP}^{50:95}$, AP^{50}, AP^{75}, and $\text{AR}^{50:95}$, and the numbers in the top-right corner represent the range of IoU, i.e., the average value over IoUs $\in [0.50 : 0.05 : 0.95]$, and when the IoU is 0.5 and 0.75, respectively. The metrics calculated based on the object area include AP_S, AP_M, AP_L, AR_S, AR_M, and AR_L, representing the performance of the detector on small objects (with an area less than 32 × 32), medium objects (with an area greater than 32 × 32 but less than 96 × 96), and large objects (with an area greater than 96 × 96), respectively.

Average precision (AP) metric indicates the detection results for all classes, and average recall (AR) metric is calculated as the maximum recall achieved with a fixed number of detections per image, averaged over object categories and IoUs $\in [0.5, 1.0]$. Both of them are computed based on the precision–recall (PR) curve.

Given an IoU threshold, the confusion matrix consisting true positive (TP), true negative (TN), false positive (FP), and false negative (FN) is determined. Then, precision and recall are computed by

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{19}$$

Furthermore, the AP for each category is computed as the area under the PR curve using the following formula:

$$\text{AP}_i = \int_0^1 P(R)\, dR. \tag{20}$$

Also, the precision of all categories AP is calculated by

$$\text{AP} = \frac{\sum_{c=1}^{N_{\text{cls}}} \text{AP}}{N_{\text{cls}}} \tag{21}$$

where N_{cls} is the total number of categories.

Likewise, AR can be formulated as follows:

$$\text{AR} = \frac{\sum_{c=1}^{N_{\text{cls}}} \text{AR}_i}{N_{\text{cls}}} \tag{22}$$

where AR_i is defined as follows:

$$\text{AR}_i = 2 \int_{0.5}^1 \text{recall}(U)\, dU \tag{23}$$

where U is the IoU thresholds $\in [0.5, 1.0]$ and recall(U) is the recall under different IoUs.

C. Experimental Setup

1) Baseline: To obtain features with clear spatial details for small objects, the proposed method's baseline is Faster R-CNN [38] with ResNet50 [47] and FPN [30]. Table I demonstrates the evaluation of the three proposed modules on this baseline.

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TABLE I
ABLATION EXPERIMENTS ON ITCVD. THE BEST RESULTS ARE MARKED IN BOLD, WHILE THE SECOND-BEST RESULTS ARE UNDERLINED

Exp. NO.	NAM	PRM	CAM	$AP^{50:95}$ /%	AP^{50} /%	AP^{75} /%	AP_S /%	AP_M /%	AP_L /%	$AR^{50:95}$ /%	AR_S /%	AR_M /%	AR_L /%	FLOPs / T	Params / M
1				53.4	83.4	63.4	30.5	56.4	35.8	62.7	44.4	65.6	39.1	0.20	41.348
2	✓			54.9	83.6	65.0	30.4	58.1	34.4	63.7	44.6	66.7	37.0	0.20	41.348
3		✓		54.7	83.6	65.3	**32.7**	57.7	35.6	63.9	44.9	66.9	40.0	0.20	41.349
4	✓	✓		55.4	<u>84.0</u>	66.2	<u>32.3</u>	58.5	35.3	64.2	**45.8**	67.2	36.1	0.20	41.349
5			✓	54.3	83.8	64.2	31.3	57.5	36.8	63.5	44.2	66.6	43.0	0.21	44.891
6	✓	✓	✓	54.9	**84.2**	65.3	**32.7**	57.9	35.3	63.6	<u>45.4</u>	66.5	37.8	0.21	44.892

TABLE II
MAGNIFICATION "k" PARAMETER SELECTION

	AP^{50} /%	AP_S /%	AP_M /%	AP_L /%	AR_S /%	AR_M /%	AR_L /%
1.5	**88.1**	11.1	47.7	53.5	22.4	**58.3**	**62.5**
2	87.9	10.2	46.2	53.7	**23.2**	56.3	62.4
3	88.0	13.2	**48.5**	**53.8**	21.4	57.6	**62.5**
4	88.0	**14.3**	46.2	53.6	22.7	56.9	62.4

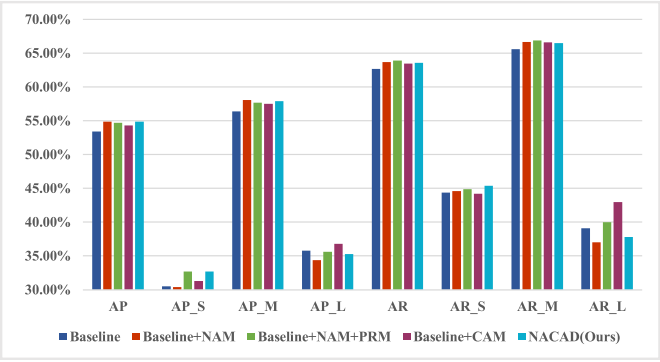


Fig. 6. Ablation experiments on ITCVD [44] (COCO format).

2) Environment: The experimental environment is PyTorch 1.7, and models are trained using the MMDetection framework[1] on Ubuntu 18.04 LTS with GeForce RTX 2080 Ti GPU.

3) Optimizing: During training, the model is optimized using stochastic gradient descent (SGD), training on a distributed setup with two GPUs and a batch size of 4. The initial learning rate is 0.01 and the training process involves 12 epochs, among which the initial learning rate is reduced by factors of 10 at the 8th and 11th epochs, with a weight decay of 0.0001 and a momentum of 0.9.

4) Parameters: The input image has a size of 1333 × 800, and Table II demonstrates that the detection performance of NACAD is affected by the magnification factor "k." Considering the results on all scales, k is finally set to 3.

D. Ablation Study

To validate the effectiveness of NAM, CAM, and PRM in our algorithm, many ablation experiments are conducted. Tables I and III show the quantitative results, while Figs. 6 and 7 illustrate the qualitative outcomes. Especially, Fig. 7 displays the visualization results of feature maps for models with different components.

1) Effectiveness of NAM: The proposed NAM is devised to mitigate the loss of HRS small objects caused by offset-sensitive problem. As demonstrated in Table I, our method, which only incorporates NAM, enhances the recall of small objects $\boldsymbol{AR_S}$ by 0.2% in comparison with the baseline, while the recall of all objects $AR_{50:95}$ is also improved by 0.9%. Because most objects in ITCVD dataset are small objects, this result is convincing. Moreover, Fig. 7(c) illustrates that the $C3$ feature map produced by FPN with NAM expands the object regions. Besides, even if features of the detected object will be disturbed when the camouflaged BBox contains other RB, the contextual information about the surrounding environment provided by NAM can mitigate this adverse impact. As shown in Table I, with the help of context information, the detection accuracy of small objects $\boldsymbol{AP_S}$ only decreases by 0.1%, rather than a more significant drop.

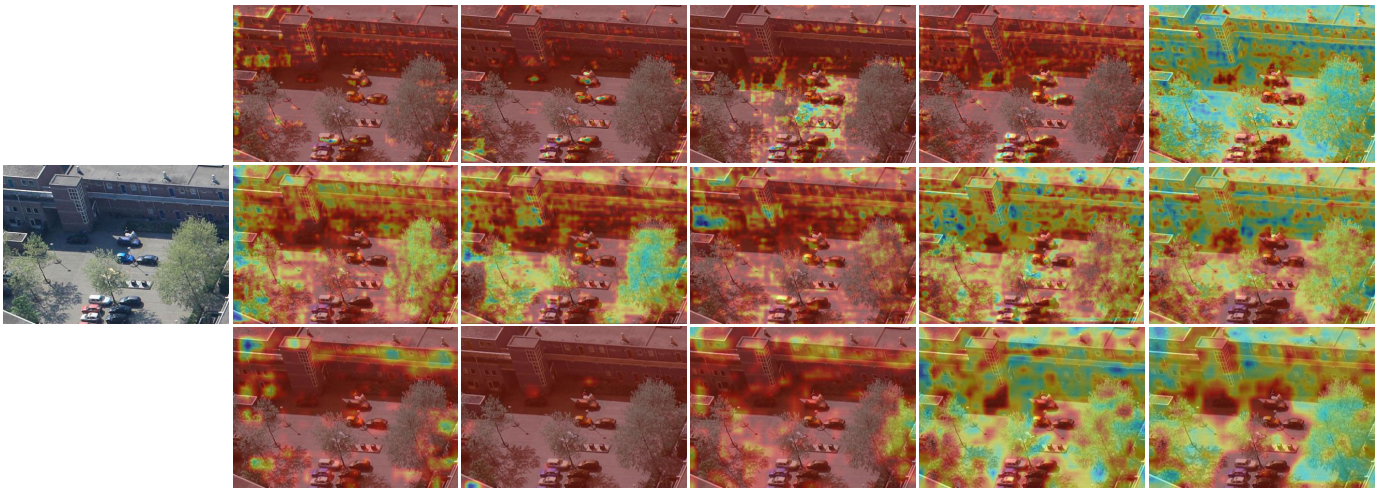
2) Effectiveness of CAM: The CAM is proposed to better highlight or differentiate HRS small objects and is compared with the baseline to show its validity. As shown in Table I, our method with only CAM can obtain 31.3% $\boldsymbol{AP_S}$ (0.8 ↑) on the ITCVD dataset. It also has the ability to increase several detection performance, especially in AP_L, which alleviates the impact from NAM on large-scale objects. Furthermore, Fig. 7(e) shows that with the assistance of contextual information, both object and nonobject regions become more distinct. With the feature maps becoming more discriminative than baseline framework, discriminant capability of the model is further enhanced, which demonstrates the effectiveness of our CAM. It has good ability to draw implicit information for small objects and extract underlying relationship of objects.

Besides, to establish the optimal configuration of CAM components, various ablation experiments are performed on the HRRSD dataset. Table III reveals that the impact of CAM on the $C3$ feature layer of FPN is superior to other layers, indicating that the multiple contextual features extracted from the $C3$ feature layer are more informative. The experimental findings also indicate that the combination of dilation rates 2, 4, 6, and 8 outperforms other combinations.

3) Effectiveness of PRM: To enhance the model's capability of reducing the interference of HRS background pure noise and separating it from π-noise, the PRM leverages spatial calibration and attention mechanism to reduce the aliasing from background and other objects. In order to confirm this, we carried out experiments by solely using PRM and by integrating PRM with NAM, respectively. From Table I, when

[1]https://github.com/open-mmlab/mmdetection

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(a) (b) (c) (d) (e) (f)

Fig. 7. Features visualization of Faster R-CNN and NACAD with three modules. (a) Input image from the ITCVD dataset. (b)–(f) First three layer feature maps of FPN ($C1$–$C3$), top to bottom.

TABLE III

EXPERIMENTAL OUTCOMES OF FPN WITH CAM ON THE HRRSD DATASET ARE PRESENTED, WHERE THE BEST RESULTS FOR EACH FEATURE LAYER ARE INDIVIDUALLY UNDERLINED, AND OVERALL BEST RESULTS ARE IN BOLD

Exp.NO.	Dilation Rates												Metrics									
	$C1$				$C2$				$C3$				$C4$				$AP^{50:95}$	AP^{50}	AP^{75}	AP_S	AP_M	AP_L	$AR^{50:95}$	AR_S	AR_M	AR_L
	2	4	6	8	2	4	6	8	2	4	6	8	2	4	6	8	/%	/%	/%	/%	/%	/%	/%	/%	/%	/%
1	✓																54.9	87.6	61.0	5.6	**48.5**	51.7	64.1	16.7	57.9	60.3
2	✓	✓															54.6	87.3	60.3	5.8	46.0	51.5	63.8	15.2	54.2	60.1
3	✓	✓	✓														54.7	87.4	60.6	8.6	47.2	51.7	63.6	**30.5**	56.3	59.9
4	✓	✓	✓	✓													55.0	87.5	60.6	11.6	44.7	51.9	64.2	24.8	55.0	60.6
5					✓												55.1	87.7	61.6	10.7	46.6	52.0	64.1	24.3	57.0	60.3
6					✓	✓											54.8	87.4	60.9	9.5	46.6	51.7	64.0	21.4	55.4	60.3
7					✓	✓	✓										54.6	87.5	60.7	6.7	45.2	51.5	63.9	29.0	56.8	60.1
8					✓	✓	✓	✓									54.4	87.6	60.0	8.0	45.7	51.5	63.6	16.2	56.3	60.0
9									✓								54.6	87.5	60.4	8.2	47.5	51.4	63.7	21.4	**58.2**	59.9
10									✓	✓							54.6	87.7	60.5	16.3	46.5	51.6	63.9	**30.5**	58.0	60.1
11									✓	✓	✓						54.7	87.6	60.4	11	46.1	51.7	63.8	12.9	55.3	60.1
12									✓	✓	✓	✓					**56.7**	**88.3**	**63.9**	**16.9**	47.4	**54.0**	**66.2**	25.0	58.0	**62.8**
13													✓				54.8	87.7	60.8	12.7	44.7	52.5	64.4	21.2	55.0	61.4
14													✓	✓			54.5	87.7	60.3	13.7	44.5	52.1	64.0	22.5	54.5	61.0
15													✓	✓	✓		54.9	87.7	61.1	9.1	46.1	52.5	64.5	16.3	56.8	61.4
16													✓	✓	✓	✓	55.0	87.6	61.4	11.6	46.5	52.6	64.5	16.5	55.9	61.4

TABLE IV

COMPARISON OF DETECTION PERFORMANCE WITH STATE-OF-THE-ART METHODS ON ITCVD

	$AP^{50:95}$ /%	AP^{50} /%	AP^{75} /%	AP_S /%	AP_M /%	AP_L /%	$AR^{50:95}$ /%	AR_S /%	AR_M /%	AR_L /%
Baseline	53.4	83.4	63.4	30.5	56.4	35.8	62.7	44.4	65.6	39.1
ATSS	52.7	84.1	61.7	30.2	55.7	**38.4**	63.5	45.3	66.3	42.2
TOOD	52.5	83.5	61.0	29.4	55.9	29.1	62.7	42.0	66.0	29.1
RetinaNet	31.1	62.1	27.7	14.1	34.5	16.5	45.6	14.4	50.5	17.8
FoveaBox	45.3	81.0	47.1	21.6	48.5	27.1	56.5	38.0	59.4	30.4
PANet	54.5	84.0	64.6	31.2	57.6	36.1	63.3	44.3	66.3	**44.8**
Double-Head R-CNN	54.3	83.6	64.8	31.4	57.1	36.0	62.9	44.9	65.7	40.0
NACAD(Ours)	**54.9**	**84.2**	**65.3**	**32.7**	**57.9**	35.3	**63.6**	**45.4**	**66.5**	37.8

adding PRM to the baseline, the improvement of $\boldsymbol{AP_S}$ on the ITCVD dataset can reach 32.7% (2.2% ↑). When integrating both PRM and NAM into the baseline, it can achieve 45.8% $\boldsymbol{AR_S}$ (1.4% ↑) with 32.3% $\mathrm{AP}_{\boldsymbol{S}}$ (1.8% ↑), simultaneously improving AP and recall performance of small objects, which alleviates the adverse impact of camouflaged BBox. Besides, the qualitative analysis is also illustrated in Fig. 7(d), for regions around small objects, the difference between object

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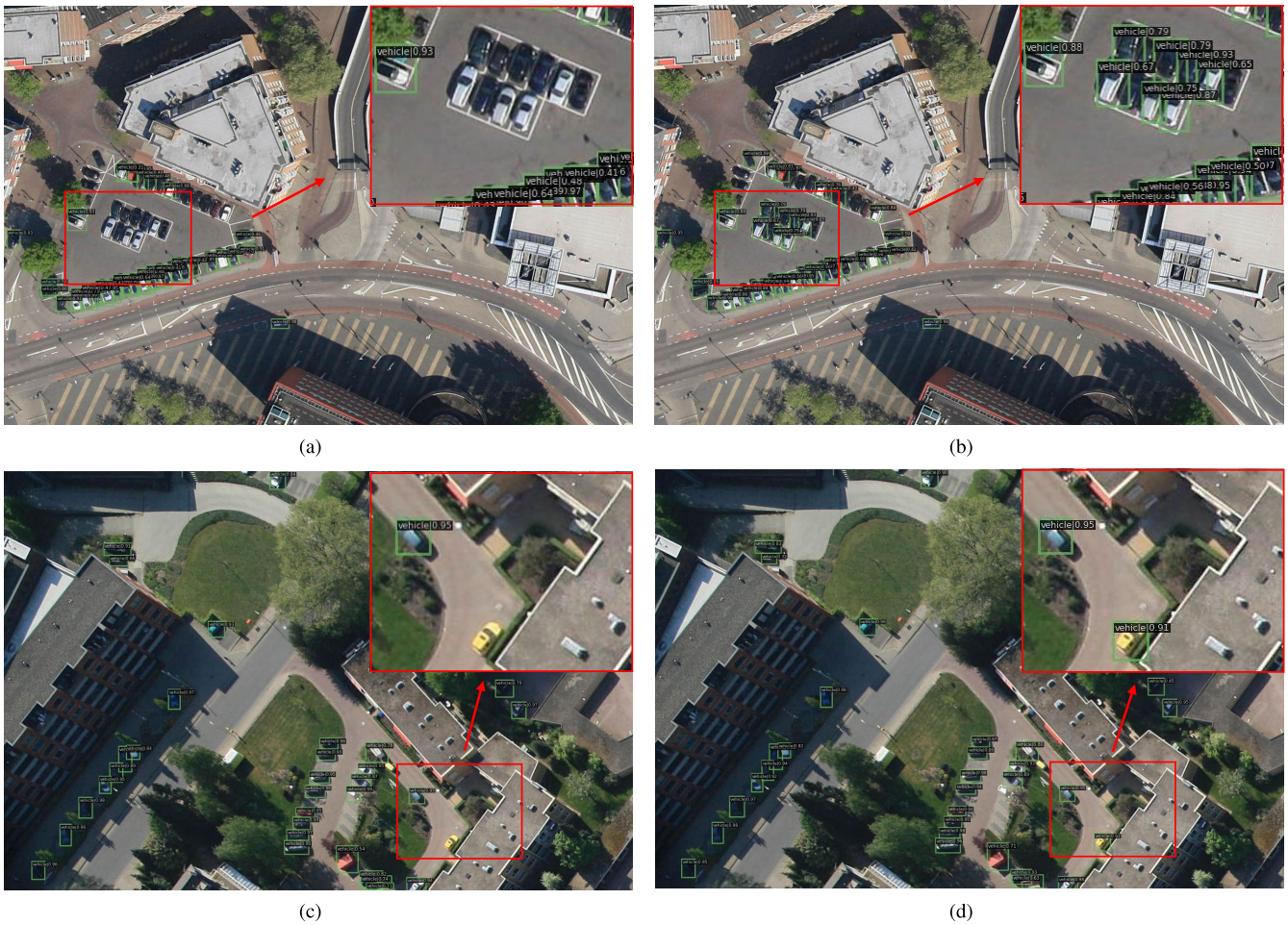


Fig. 8. Detection Results of Faster R-CNN and NACAD on the ITCVD test split. (a) and (c) Results of Faster R-CNN. (b) and (d) Results of NACAD.

features and background features is more pronounced, which is conducive for precise object localization.

The analysis presented above demonstrates the effectiveness of each individual module. To reinforce the credibility of combining them, the experimental results are analyzed from an overall point of view. Table I demonstrates that our NACAD achieves superior performance in terms of $\boldsymbol{AP_S}$ compared with a single module, with a value of 32.7%. In addition, our approach ranks second in terms of $\boldsymbol{AR_S}$, following only the combination of NAM and PRM. Meanwhile, the NACAD performs best on the AP_{50}, indicating that our method can be well applied to the real world. In addition, as shown in Fig. 6, for the ITCVD dataset with only a single category of small objects, the proposed NACAD advances the terms of $\boldsymbol{AR_S}$ and $\boldsymbol{AP_S}$ greatly. Moreover, Fig. 7(f) reveals that our algorithm produces noticeably superior feature maps compared with other methods, indicating our approach possesses discriminative description capabilities for potential object features, as well as higher positioning accuracy for small objects.

E. Contrast Experiments

To showcase the efficacy, our approach is compared with several advanced methods on two publicly available remote sensing datasets, namely, ITCVD and HRRSD. These advanced algorithms include one-stage detectors: ATSS [48], TOOD [49], FoveaBox [50], RetinaNet [22], and guided anchoring [51]; and two-stage detectors: Faster R-CNN, Libra R-CNN [52], PANet [53], and Double-Head R-CNN [54]. Tables IV and V present the test results of the detectors, which use ResNet-50 as backbone network and are reimplementations of MMDetection [55].

1) Results on ITCVD: To evaluate the robustness of our NACAD, contrast experiments are performed on the ITCVD dataset, which is a challenging benchmark for detecting small objects. As shown in Table IV, our NACAD is highly competitive compared with other methods, achieving the most advanced performance in several indices (indicated in bold). The largest improvement in $\boldsymbol{AP_S}$, AP_M, and AP_L is observed for small objects, with a significant increase of 2.2%. This result confirms that the NACAD has significant advantages in detecting small objects. Furthermore, compared with ATSS and PANet, our method performs a little worse in terms of AP_L and AR_L on ITCVD, but objectively speaking, its detection accuracy on large objects is still considerable. We observe that the single-class small object dataset ITCVD has a test set with 42.0% of object areas smaller than 32×32 and only 2.3% of object areas larger than 96×96. This indicates that the ITCVD dataset mainly proves the effectiveness of our method on small objects. Besides, Fig. 8 provides additional evidence

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TABLE V
COMPARISON OF DETECTION PERFORMANCE WITH STATE-OF-THE-ART METHODS ON HRRSD

	$AP^{50:95}$ /%	AP^{50} /%	AP^{75} /%	AP_S /%	AP_M /%	AP_L /%	$AR^{50:95}$ /%	AR_S /%	AR_M /%	AR_L /%
Baseline	56.6	88.5	63.9	5.7	47.9	53.7	65.5	8.1	59.0	61.6
ATSS	56.8	87.0	**64.4**	9.5	46.2	**54.1**	**68.8**	17.0	**63.6**	**65.3**
FoveaBox	52.4	83.6	57.9	6.5	43.6	49.5	65.1	21.2	60.8	61.6
RetinaNet	52.0	83.0	56.9	15.8	42.0	49.1	65.5	29.1	61.6	62.0
Libra R-CNN	56.3	87.5	63.5	11.4	46.0	53.9	65.4	26.5	58.1	62.1
Guided Anchoring	55.5	86.7	61.9	13.0	45.4	52.6	65.9	**30.0**	60.3	62.4
Double-Head R-CNN	56.3	88.0	63.1	12.2	48.3	53.6	64.6	20.1	58.7	61.4
NACAD(Ours)	**57.0**	**88.6**	64.2	**16.4**	**48.5**	**54.1**	66.1	23.5	59.5	62.7



Fig. 9. Proposed method's detection results on the HRRSD test split are displayed, with confidence percentages indicated on the detected objects.

of the effectiveness of NACAD in detecting small objects. The baseline and NACAD network's detection results are presented in the first and second columns, respectively. As we can see, several vehicles are missed in Fig. 8(a) and (c). However, they can be detected by our proposed method, as shown in Fig. 8(b) and (d). Thus, we can conclude that introducing π-noise reasonably and considering contextual information can improve the recall performance and keep satisfactory accuracy.

2) Results on HRRSD: There are a total of 13 categories in HRRSD dataset, including various small objects categories, such as ships, vehicles, storage tanks, and tennis courts, making it challenging for a general detector to perform well on it. As shown in Table V, various methods' performance is quantitatively reported using the MS COCO metrics. By analyzing the results of different on small objects, our method reaches 16.4% $\boldsymbol{AP_S}$ on the HRRSD dataset, 10.7% more than baseline, achieving the best performance among competitors. Moreover, the proposed modules effectively improve the precision of object detection, as evidenced by the improvements of 0.4%, 0.1%, 10.7%, 0.6%, and 0.4% in $AP^{50:95}$, AP^{50}, $\boldsymbol{AP_S}$, AP_M, and AP_L detection results, respectively. Also, the recall performance of different objects $\boldsymbol{AR_S}$, AR_M, and AR_L is also increased by 15.4%, 0.5%, and 1.1%, respectively. The results demonstrate that our method can enhance the detection performance of small objects without compromising the detection of large objects. This further validates the effectiveness of our method in practical applications. Besides, as shown in Fig. 9, our NACAD has promising results on 13 categories, which fully gives expression to the effectiveness to detection multiple small objects categories in real world.

3) Complexity and Speed Comparison: In addition to the above analysis, we have also conducted a comparison of the complexity, speed, and detection accuracy on small objects between our method and other approaches. As presented in Fig. 10, the NACAD demonstrates clear superiority in terms of speed and detection performance on small objects, while its complexity remains comparable to that of two-stage detectors,

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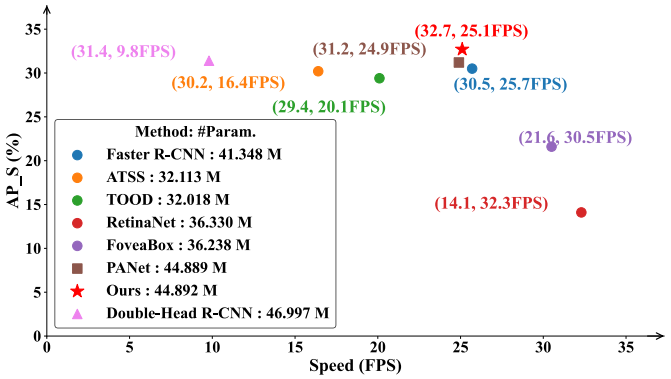


Fig. 10. Detection accuracy on small objects (AP_S) versus speed (FPS) comparison on the ITCVD test set.

such as PANet and Double-Head R-CNN. Beyond this, our method outperforms one-stage algorithms in terms of speed and accuracy on small objects. Specifically, it surpasses ATSS and TOOD in terms of speed and achieves higher detection accuracy for small objects compared with RetinaNet and FoveaBox. In addition, although it can be observed from Table I that the increased model complexity of NACAD primarily stems from the CAM module, experiments demonstrate that incorporating this module yields optimal results across various metrics. Notably, NAM does not contribute to increased complexity, and PRM only adds a mere 0.001M to the Params, highlighting the efficiency and simplicity of these two modules.

V. Conclusion

In this article, a novel detector NACAD that includes three meticulously designed modules is proposed based on Faster R-CNN, which can effectively alleviate the challenges in HRS small object detection. First, with the aim of avoiding missing small objects, our simple but effective NAM can obtain impressive recall improvement for small objects. Furthermore, the CAM boosts the discrimination of HRS small objects by injecting contextual representations effectively. Finally, the PRM filters pure noise in background information around small objects using spatial calibration to improve the anti-interference ability of the model. Considerable experiments are performed on two challenging datasets (including small objects, such as ships, vehicles, and storage tanks), which indicates the remarkable performance of our proposed NACAD.

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